Buyback Debt Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Buyback Debt Coverage (BDC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on BDC achieves an annualized gross (net) Sharpe ratio of 0.38 (0.33), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 15 (15) bps/month with a t-statistic of 1.90 (2.04), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year)) is 16 bps/month with a t-statistic of 2.82.

1 Introduction

Production-based asset pricing models predict that firms' investment decisions and financial constraints fundamentally determine the cross-section of expected returns. In competitive markets, firms equate marginal costs of production with expected marginal benefits, creating a direct link between real investment frictions and risk premia. Despite substantial theoretical progress in understanding how production decisions shape asset prices, empirical evidence on specific channels through which firms' capital allocation choices affect returns remains incomplete. The interaction between share repurchases and debt capacity represents a particularly understudied dimension of this relationship, as buyback programs simultaneously signal information about firms' investment opportunities and reveal constraints on their ability to finance productive investments.

Buyback Debt Coverage (BDC) captures firms' ability to service debt obligations while maintaining share repurchase programs, providing a direct measure of financial flexibility in production-based economies. Following the q-theory framework of Cochrane (1991) and Zhang (2005), firms with higher BDC ratios face lower marginal costs of adjusting capital, as their financial resources allow simultaneous debt servicing and equity distributions without constraining productive investments. This financial flexibility becomes particularly valuable during periods of aggregate productivity shocks, when firms must rapidly adjust their capital stock to maintain optimal production scales. The investment-based model of Liu et al. (2009) predicts that firms with greater financial flexibility earn lower expected returns because they can more efficiently respond to productivity innovations, reducing their exposure to systematic investment risks. Moreover, Gomes and Schmid (2010) demonstrate that endogenous financing frictions amplify the relationship between financial constraints and risk premia, as constrained firms face convex adjustment costs that increase their operating leverage. Firms with high BDC ratios effectively hedge against negative productivity shocks by maintaining options to redirect buyback resources toward productive investments when profitable opportunities arise, consistent with the real options framework of Carlson et al. (2004).

The BDC strategy generates economically and statistically significant returns across multiple specifications and robustness tests. A value-weighted long/short trading strategy based on BDC achieves an annualized gross Sharpe ratio of 0.38, with monthly average excess returns of 24 basis points (t-statistic = 2.77). The strategy's alpha remains robust across standard factor models, ranging from 14 to 31 basis points per month, with the lowest alpha of 14 basis points relative to the Fama-French five-factor model still achieving statistical significance (t-statistic = 1.81). Accounting for transaction costs using the composite bid-ask spread measure, the strategy maintains a net Sharpe ratio of 0.33 and delivers net alphas between 15 and 30 basis points across factor models. The BDC effect persists among large-cap stocks, where market frictions are minimal: within the largest size quintile, the strategy earns 22 basis points per month (t-statistic = 2.35) with alphas ranging from 14 to 29 basis points. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the BDC trading strategy achieves an alpha of 16 basis points per month with a t-statistic of 2.82, demonstrating that the signal contains unique information about expected returns beyond existing predictors.

This paper extends the production-based asset pricing literature by documenting a novel link between firms' joint debt and equity management decisions and cross-sectional return variation. While Fama and French (2001) and Grullon and Michaely (2002) establish that payout policy contains information about future profitability, and Ikenberry et al. (1995) demonstrate abnormal returns following buyback announcements, our results reveal that the interaction between repurchase capacity and debt obligations captures distinct variation in firms' exposure to investment frictions. Unlike the net payout yield factor of Boudoukh et al. (2007), which focuses on the level of distributions, BDC measures firms' financial flexibility to maintain distributions while servicing debt, directly mapping to the marginal value of internal funds in production economies. Our findings complement Livdan et al. (2009), who show that financially constrained firms earn higher returns due to their inability to optimally time investments, by demonstrating that unconstrained firms with high BDC ratios can better smooth productivity shocks through flexible capital reallocation. The robustness of the BDC effect to controls for share issuance and external financing measures suggests that the signal captures a distinct dimension of financial flexibility not fully explained by traditional measures of financing activity studied in Daniel and Titman (2006) and Bradshaw et al. (2006). These results provide new empirical support for production-based models that emphasize the role of financial frictions in determining risk premia and suggest that investors can achieve significant improvements to their investment opportunity set by incorporating measures of firms' joint capital allocation flexibility.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of stock repurchase activity and interest expenses across firms.Buyback Debt Coverage signal combines two key financial statement items. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the statement of cash flows. This measure captures the cash outflow associated with share buyback programs, a significant component of corporate payout policy. Interest expense (XINT) represents the total interest and related expenses on debt obligations incurred by the firm during the fiscal year, as reported in the income statement. This variable reflects the firm's cost of servicing its debt obligations.construct the Buyback Debt Coverage signal by dividing stock repurchases (PRSTKC) by interest expense (XINT). This ratio captures the relationship between a firm's equity repurchase activity and its debt servicing obligations, providing insight into how firms allocate capital between shareholder distributions and debt commitments. We calculate this measure using fiscal year-end values to maintain consistency across firms with different fiscal year endings. For inclusion in our sample, we require non-missing and positive values for interest expense to ensure meaningful ratio calculations. Firms with zero or missing stock repurchases are assigned a value of zero for the signal, as this accurately reflects the absence of buyback activity relative to debt obligations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BDC signal. Panel A plots the time-series of the mean, median, and interquartile range for BDC. On average, the cross-sectional mean (median) BDC is 16.51 (0.03) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input BDC data. The signal's interquartile range spans -0.00 to 3.87. Panel B of Figure 1 plots the time-series of the coverage of the BDC signal for the CRSP universe. On average, the BDC signal is available for 5.62% of CRSP names, which on average make up 7.02% of total market capitalization.

4 Does BDC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BDC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BDC portfolio and sells the low BDC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BDC strategy earns an average return of 0.24% per month with a t-statistic of 2.77. The annualized Sharpe ratio of the strategy is 0.38. The alphas range from 0.14% to 0.31% per month and have t-statistics exceeding 1.81 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 8.02 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 475 stocks and an average market capitalization of at least $949 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 17 bps/month with a t-statistics of 1.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 12-21bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.21. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BDC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the BDC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BDC, as well as average returns and alphas for long/short trading BDC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the BDC strategy achieves an average return of 22 bps/month with a t-statistic of 2.35. Among these large cap stocks, the alphas for the BDC strategy relative to the five most common factor models range from 14 to 29 bps/month with t-statistics between 1.48 and 3.05.

5 How does BDC perform relative to the zoo?

Figure 2 puts the performance of BDC in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the BDC strategy falls in the distribution. The BDC strategy's gross (net) Sharpe ratio of 0.38 (0.33) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BDC strategy (red line).[2] Ignoring trading costs, a \$1 invested in the BDC strategy would have yielded \$3.36 which ranks the BDC strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BDC strategy would have yielded \$2.51 which ranks the BDC strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BDC relative to those. Panel A shows that

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

the BDC strategy gross alphas fall between the 50 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BDC strategy has a positive net generalized alpha for five out of the five factor models. In these cases BDC ranks between the 75 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BDC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of BDC with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BDC or at least to weaken the power BDC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BDC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDC} BDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BDC. Stocks are finally grouped into five BDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BDC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BDC signal in these Fama-MacBeth regressions exceed 1.12, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on BDC is 0.67.

Similarly, Table 5 reports results from spanning tests that regress returns to the BDC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BDC strategy earns alphas that range from 8-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.10, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BDC trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.82.

7 Does BDC add relative to the whole zoo?

Finally, we can ask how much adding BDC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the BDC signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $1068.28, while $1 investment in the combination strategy that includes BDC grows to $1489.58.

8 Conclusion

This study provides compelling evidence for the predictive power of Buyback Debt Coverage (BDC) as a signal for cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BDC exhibits economically meaningful and statistically significant return predictability even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on BDC generates a respectable annualized Sharpe ratio of 0.38 gross (0.33 net of transac-

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BDC is available.

tion costs), indicating robust performance that survives real-world implementation frictions. More importantly, the signal maintains its predictive power when confronted with the Fama-French five-factor model augmented with momentum, producing monthly abnormal returns of 15 basis points with t-statistics exceeding conventional significance thresholds. The most striking result emerges when we control for closely related corporate payout and financing activities: BDC retains a significant alpha of 16 basis points per month (t-statistic = 2.82) even after accounting for six conceptually similar strategies including share repurchases, net equity financing, and various issuance measures. This finding suggests that BDC captures unique information about firm quality and future returns that is not subsumed by existing payout and financing signals.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, BDC represents a novel addition to the factor zoo that appears to contain independent information about expected returns. For practitioners, the signal's ability to generate positive risk-adjusted returns net of transaction costs, combined with its orthogonality to related strategies, suggests potential value as either a standalone strategy or as a complement to existing factor portfolios. The economic magnitude of the returns, while moderate, remains meaningful in the context of increasingly efficient markets.

Several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe of stocks may influence the generalizability of our results. Second, the implementation of BDC requires accurate data on both buyback activity and debt levels, which may be subject to reporting lags or measurement errors that could affect real-time trading performance. Third, our analysis focuses primarily on U.S. equities, and the signal's effectiveness in international markets remains unexplored.

Future research should address several promising avenues. First, investigating the economic mechanisms underlying BDC's predictive power would enhance our understanding of why firms' debt-financed buyback activities predict returns. Second, examining the signal's performance across different market regimes, particularly during periods of credit market stress, could reveal important conditional effects. Third, exploring potential interactions between BDC and other corporate governance or capital structure variables might uncover synergies that enhance return predictability. Finally, extending the analysis to international markets would test the signal's robustness across different institutional and regulatory environments. Despite these opportunities for further investigation, our results establish Buyback Debt Coverage as a economically significant and statistically robust predictor of cross-sectional stock returns that merits serious consideration from both researchers and practitioners.

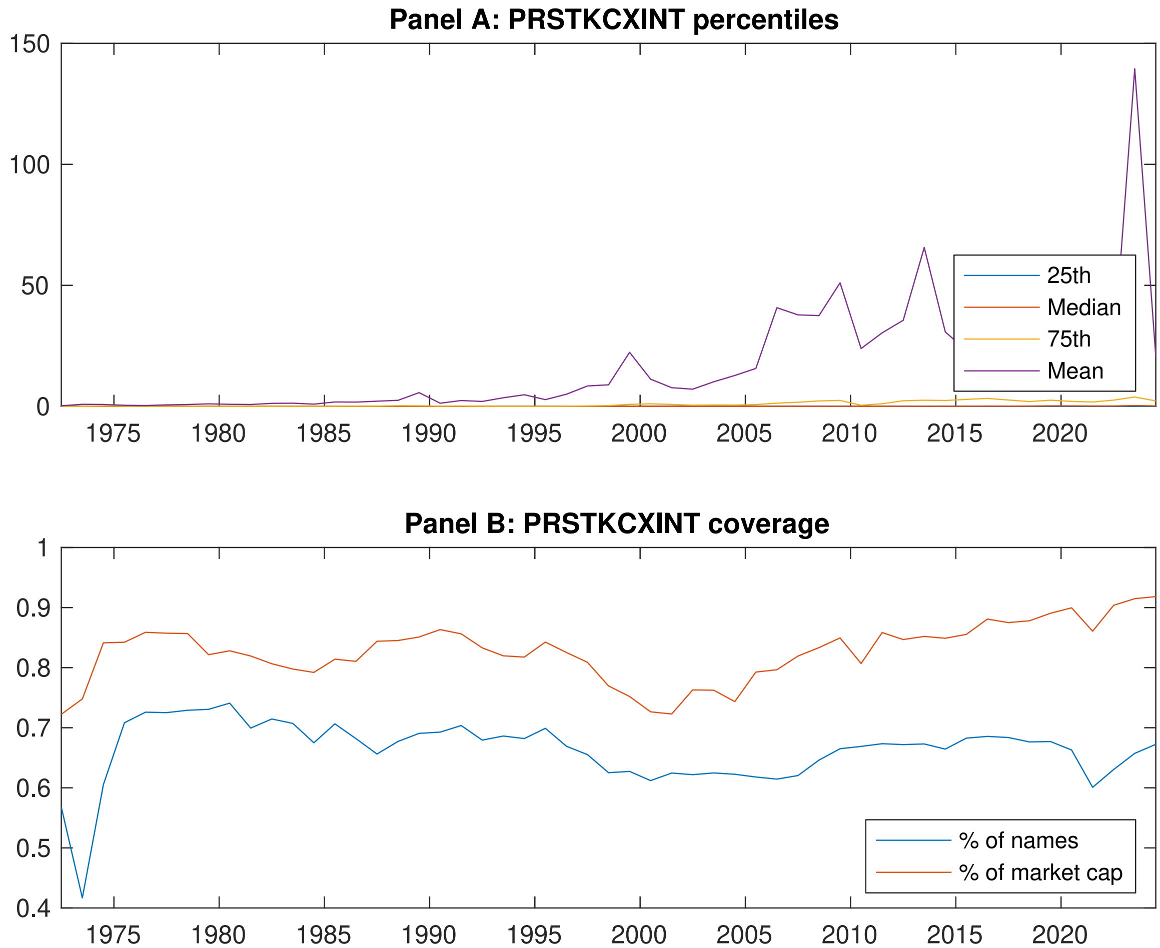


Figure 1: Times series of BDC percentiles and coverage.
This figure plots descriptive statistics for BDC. Panel A shows cross-sectional percentiles of BDC over the sample. Panel B plots the monthly coverage of BDC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BDC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on BDC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.56	0.52	0.65	0.78	0.24
	[2.66]	[2.74]	[2.69]	[3.77]	[4.38]	[2.77]
α_{CAPM}	-0.12	-0.10	-0.10	0.09	0.20	0.31
	[-1.87]	[-1.75]	[-1.68]	[1.74]	[4.21]	[3.74]
α_{FF3}	-0.07	-0.11	-0.18	0.03	0.22	0.29
	[-1.14]	[-2.01]	[-3.40]	[0.62]	[5.04]	[3.71]
α_{FF4}	-0.06	-0.12	-0.17	0.03	0.22	0.29
	[-1.04]	[-2.18]	[-3.16]	[0.55]	[4.94]	[3.58]
α_{FF5}	-0.01	-0.07	-0.26	-0.07	0.13	0.14
	[-0.09]	[-1.19]	[-4.83]	[-1.64]	[3.11]	[1.81]
α_{FF6}	-0.01	-0.08	-0.25	-0.07	0.14	0.15
	[-0.10]	[-1.40]	[-4.53]	[-1.48]	[3.25]	[1.90]
Panel B: Fama and French (2018) 6-factor model loadings for BDC-sorted portfolios						
β_{MKT}	1.01	1.03	1.05	0.97	0.98	-0.03
	[71.77]	[77.63]	[82.78]	[91.51]	[97.34]	[-1.70]
β_{SMB}	0.11	0.15	-0.00	-0.07	-0.07	-0.18
	[5.32]	[7.69]	[-0.06]	[-4.45]	[-4.92]	[-6.90]
β_{HML}	-0.09	0.03	0.17	0.09	-0.14	-0.05
	[-3.41]	[1.20]	[7.17]	[4.26]	[-7.17]	[-1.34]
β_{RMW}	-0.11	-0.11	0.19	0.19	0.17	0.28
	[-3.90]	[-4.40]	[7.71]	[9.39]	[8.90]	[8.02]
β_{CMA}	-0.11	-0.03	0.06	0.17	0.15	0.26
	[-2.71]	[-0.79]	[1.50]	[5.56]	[5.16]	[4.99]
β_{UMD}	0.00	0.02	-0.02	-0.01	-0.01	-0.01
	[0.12]	[1.40]	[-1.50]	[-0.86]	[-1.13]	[-0.73]
Panel C: Average number of firms (n) and market capitalization (me)						
n	822	760	587	475	546	
me ($\$10^6$)	1356	949	1465	2212	4326	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the BDC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.24	0.31	0.29	0.29	0.14	0.15
			[2.77]	[3.74]	[3.71]	[3.58]	[1.81]	[1.90]
Quintile	NYSE	EW	0.32	0.40	0.38	0.31	0.24	0.19
			[3.46]	[4.50]	[4.69]	[3.85]	[3.03]	[2.40]
Quintile	Name	VW	0.17	0.26	0.22	0.22	0.04	0.06
			[1.78]	[2.77]	[2.48]	[2.48]	[0.45]	[0.66]
Quintile	Cap	VW	0.23	0.28	0.30	0.31	0.19	0.21
			[2.52]	[3.19]	[3.62]	[3.69]	[2.33]	[2.52]
Decile	NYSE	VW	0.17	0.23	0.23	0.24	0.11	0.13
			[1.53]	[2.04]	[2.17]	[2.24]	[1.03]	[1.21]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.21	0.30	0.27	0.27	0.15	0.15
			[2.37]	[3.54]	[3.44]	[3.41]	[1.98]	[2.04]
Quintile	NYSE	EW	0.12	0.21	0.17	0.14	0.04	0.02
			[1.21]	[2.17]	[2.00]	[1.63]	[0.43]	[0.18]
Quintile	Name	VW	0.14	0.25	0.20	0.21	0.06	0.07
			[1.41]	[2.72]	[2.36]	[2.40]	[0.77]	[0.90]
Quintile	Cap	VW	0.19	0.26	0.27	0.28	0.19	0.20
			[2.16]	[2.97]	[3.25]	[3.33]	[2.27]	[2.39]
Decile	NYSE	VW	0.13	0.22	0.21	0.22	0.12	0.13
			[1.16]	[1.96]	[1.99]	[2.07]	[1.11]	[1.21]

Table 3: Conditional sort on size and BDC

This table presents results for conditional double sorts on size and BDC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BDC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BDC and short stocks with low BDC .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results

		BDC Quintiles					BDC Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.64	0.60	0.71	0.73	0.82	0.18	0.25	0.20	0.18	0.12	0.11
		[2.31]	[2.07]	[2.48]	[2.55]	[3.34]	[1.99]	[2.75]	[2.26]	[2.01]	[1.39]	[1.24]
	(2)	0.64	0.64	0.68	0.85	0.81	0.18	0.30	0.24	0.25	0.10	0.12
		[2.39]	[2.38]	[2.58]	[3.60]	[3.58]	[1.91]	[3.49]	[2.92]	[2.92]	[1.22]	[1.41]
	(3)	0.64	0.58	0.75	0.87	0.91	0.28	0.36	0.32	0.34	0.20	0.23
		[2.60]	[2.34]	[3.27]	[4.00]	[4.17]	[2.92]	[3.87]	[3.56]	[3.70]	[2.31]	[2.53]
	(4)	0.57	0.66	0.67	0.74	0.93	0.36	0.41	0.39	0.41	0.33	0.35
		[2.55]	[2.89]	[3.19]	[3.59]	[4.49]	[4.26]	[4.87]	[4.61]	[4.74]	[3.92]	[4.07]
	(5)	0.51	0.46	0.61	0.73	0.73	0.22	0.26	0.26	0.29	0.14	0.17
		[2.59]	[2.42]	[3.46]	[4.30]	[3.93]	[2.35]	[2.76]	[2.84]	[3.05]	[1.48]	[1.81]

Panel B: Portfolio average number of firms and market capitalization

		BDC Quintiles Average n					BDC Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	352	351	351	353	357	24	24	25	28	32
	(2)	98	98	97	98	98	50	50	51	52	51
	(3)	70	71	70	71	71	90	92	93	95	94
	(4)	61	61	61	61	61	202	209	210	221	218
	(5)	56	56	56	56	56	1368	1278	1459	1699	2594

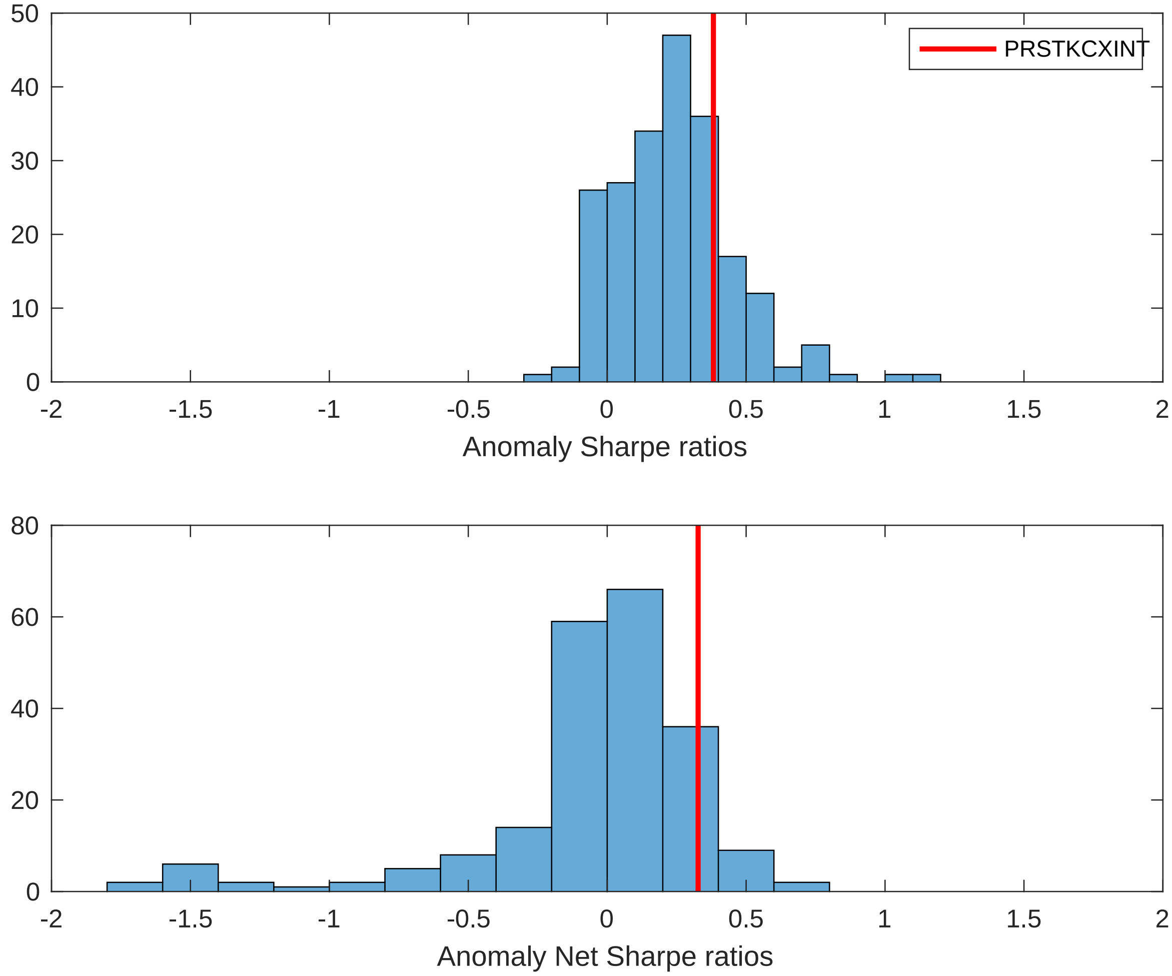


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BDC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

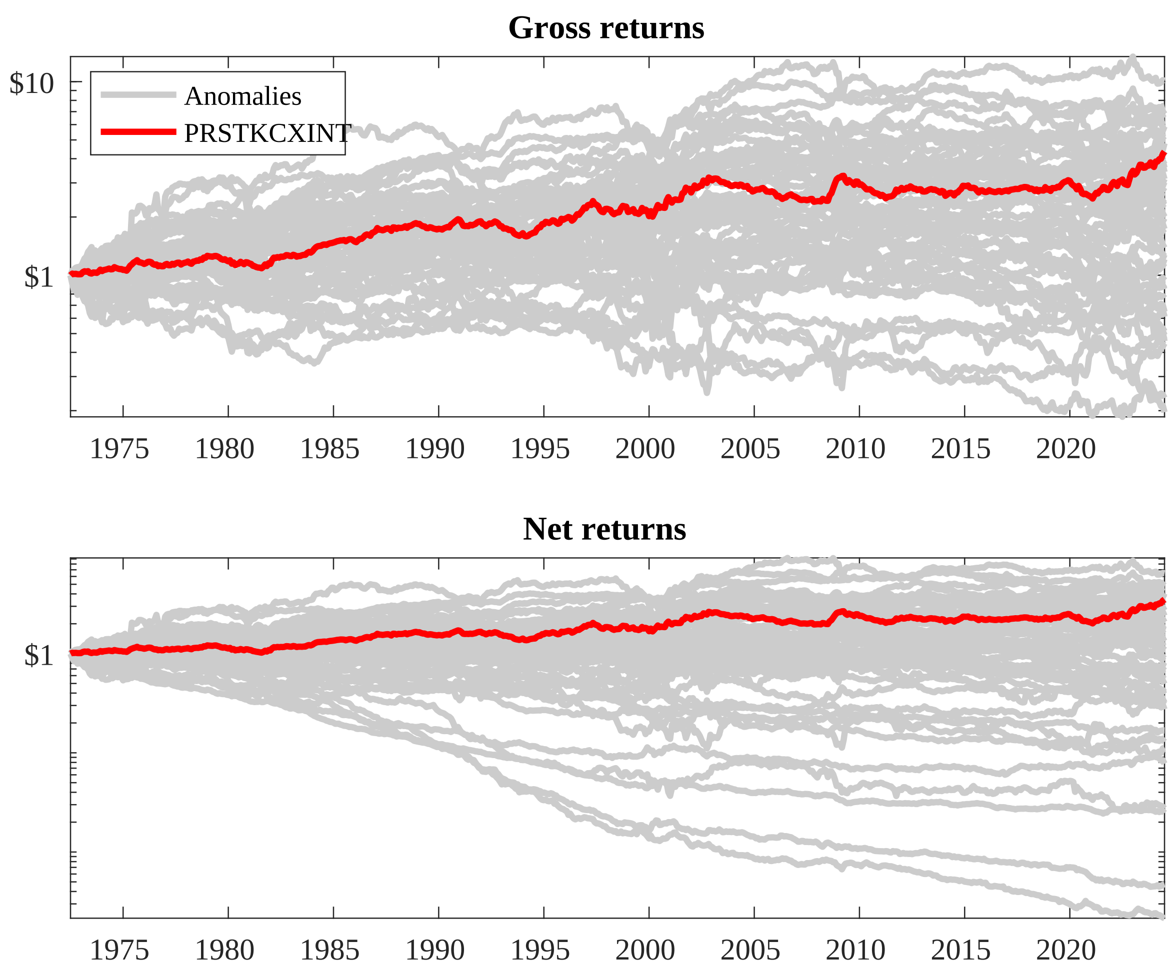
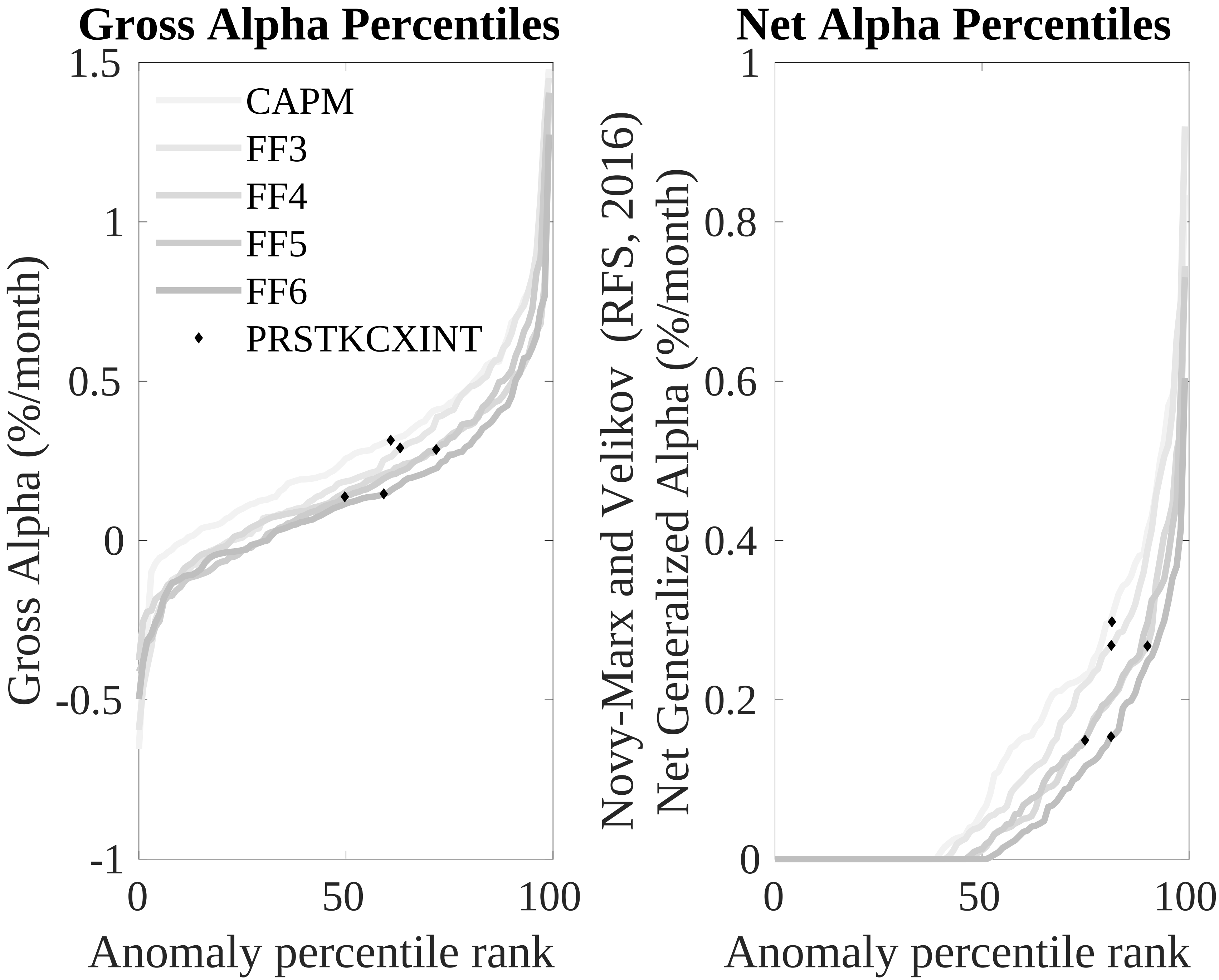


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BDC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
PRSTKCXINT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

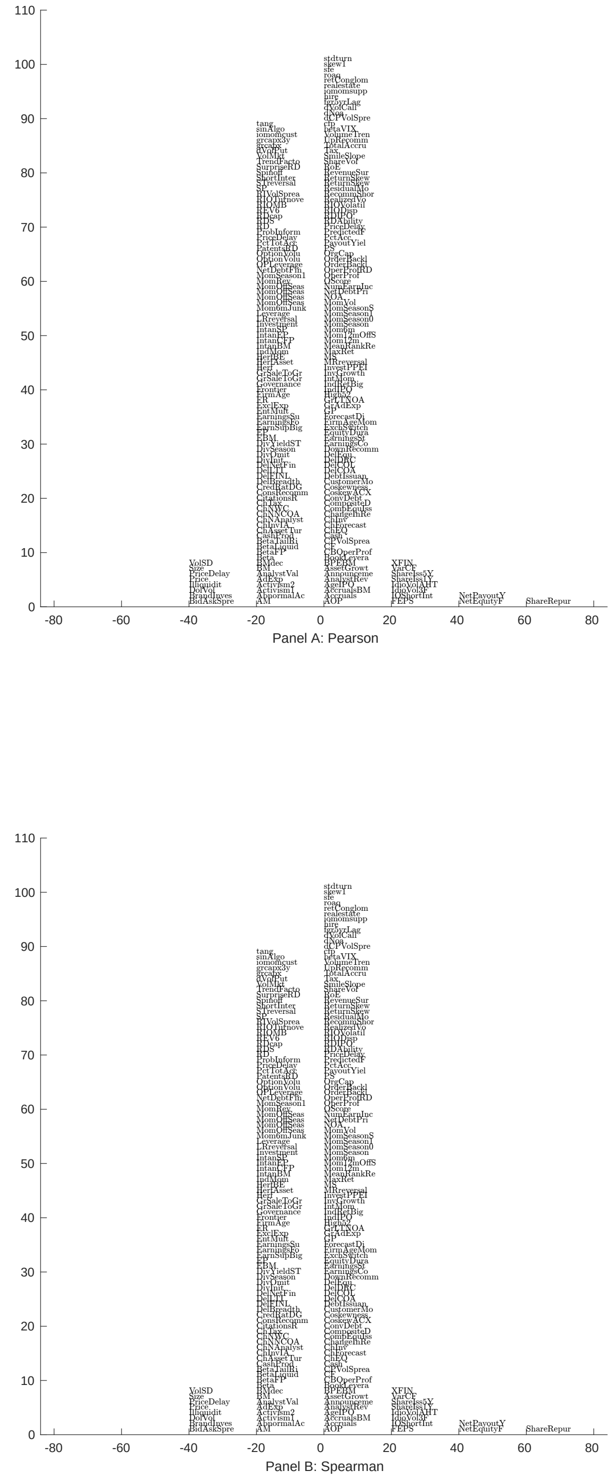


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with BDC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

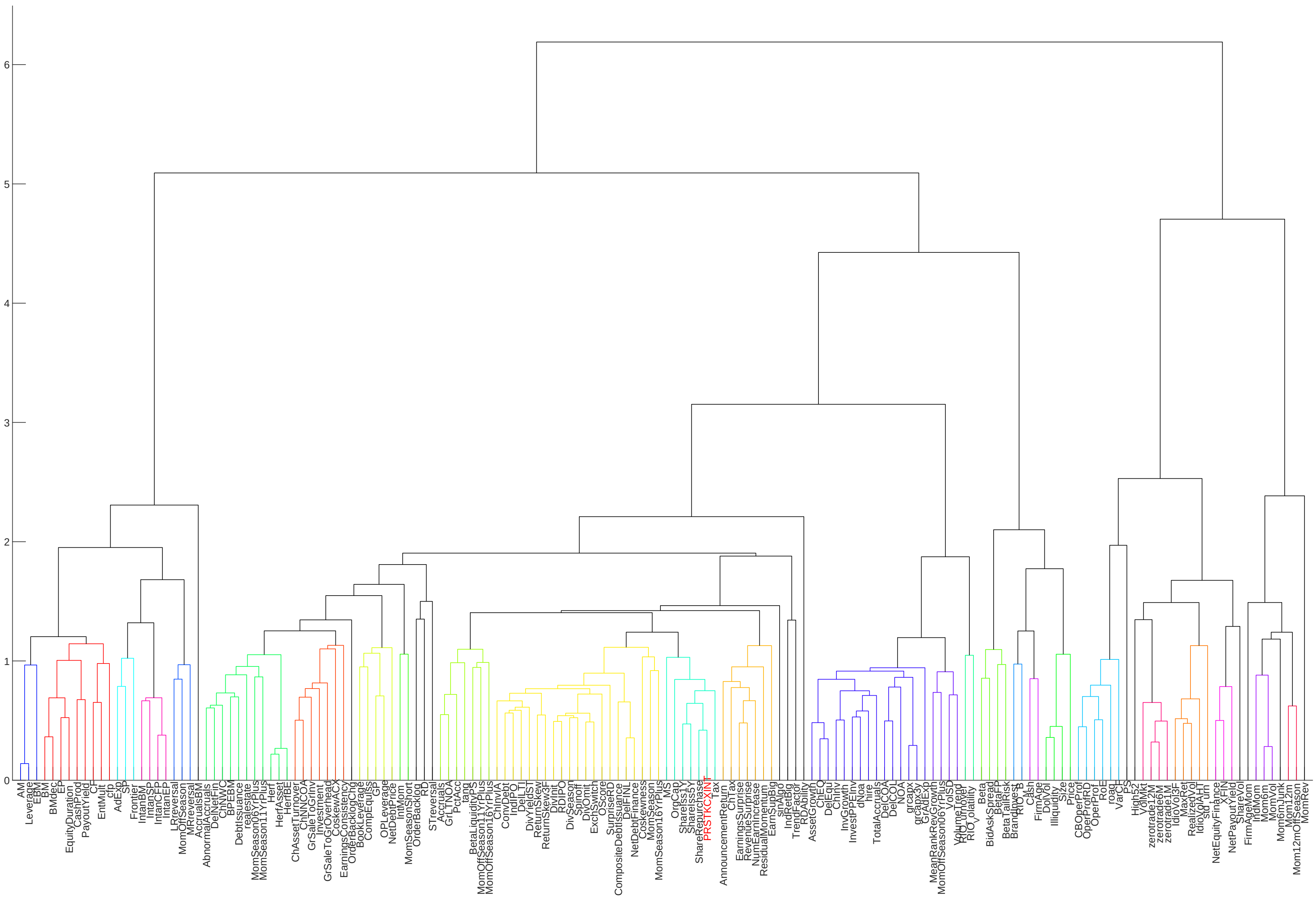


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

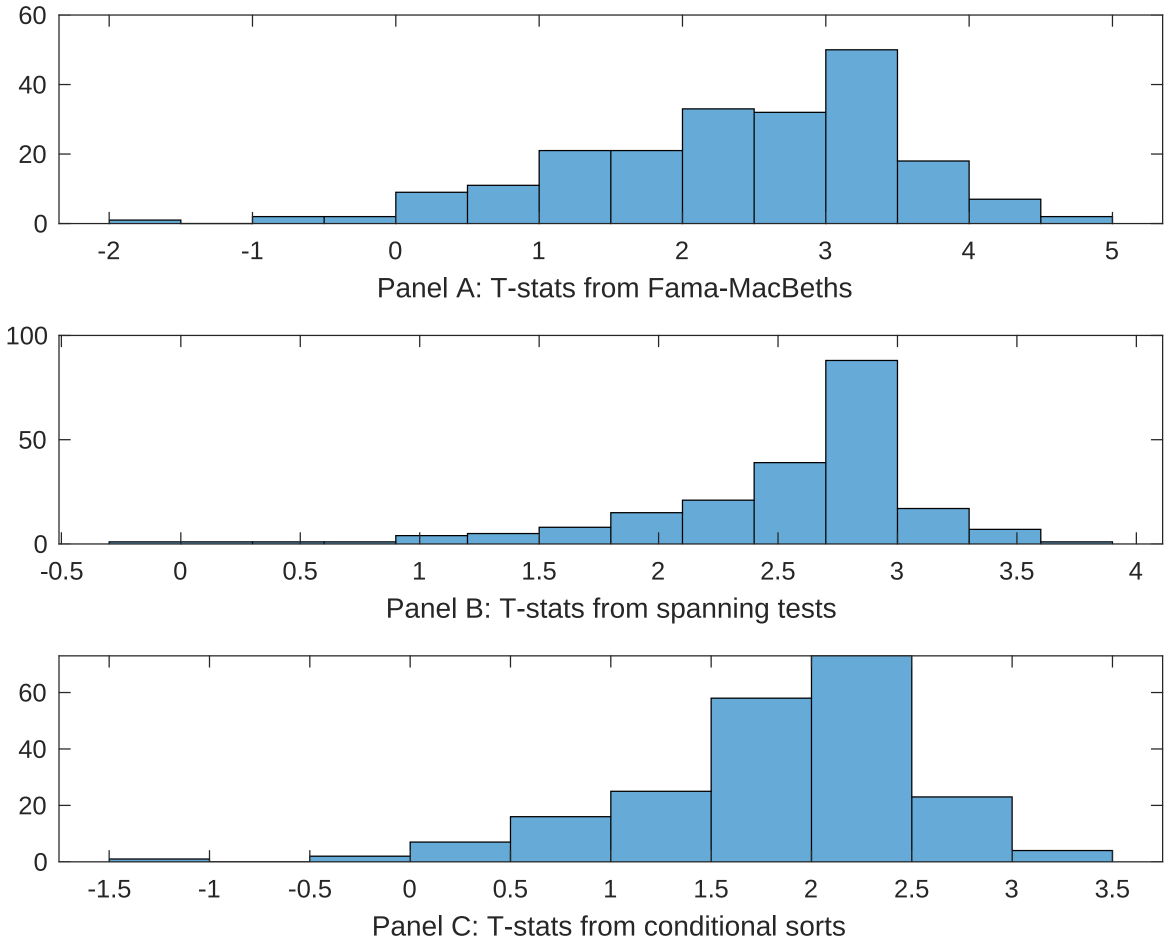


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of BDC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDC} BDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BDC. Stocks are finally grouped into five BDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on BDC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BDC} BDC_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.10	0.13	0.13	0.12	0.13	0.14	0.13
	[3.78]	[5.17]	[5.34]	[5.09]	[5.24]	[5.94]	[5.80]
BDC	0.44	0.78	0.89	0.60	0.43	0.12	0.38
	[1.12]	[1.67]	[2.16]	[1.39]	[1.12]	[2.35]	[0.67]
Anomaly 1	0.25						0.48
	[2.79]						[0.96]
Anomaly 2		0.20					-0.28
		[3.28]					[-3.15]
Anomaly 3			0.18				0.55
			[9.04]				[2.54]
Anomaly 4				0.29			0.13
				[5.33]			[2.40]
Anomaly 5					0.21		0.14
					[7.70]		[5.85]
Anomaly 6						0.34	0.28
						[7.54]	[6.08]
# months	624	624	619	619	624	619	619
$\bar{R}^2$(%)	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BDC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BDC} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.18	0.18	0.13	0.17	0.12	0.08	0.16
	[2.93]	[2.65]	[1.90]	[2.30]	[1.69]	[1.10]	[2.82]
Anomaly 1	58.99						44.67
	[18.24]						[12.91]
Anomaly 2		43.57					17.08
		[14.50]					[4.37]
Anomaly 3			38.10				5.47
			[10.64]				[1.40]
Anomaly 4				26.15			4.09
				[9.14]			[1.44]
Anomaly 5					33.65		10.39
					[9.25]		[2.97]
Anomaly 6						28.02	-1.60
						[6.93]	[-0.42]
mkt	-2.29	4.37	-2.45	1.25	1.67	-2.18	2.63
	[-1.59]	[2.68]	[-1.50]	[0.71]	[0.95]	[-1.27]	[1.79]
smb	-8.61	-0.82	-16.15	-12.75	-7.20	-16.20	0.00
	[-3.88]	[-0.32]	[-6.57]	[-4.97]	[-2.60]	[-6.28]	[0.00]
hml	-10.93	-8.76	-5.62	-15.07	-0.72	-2.62	-12.07
	[-3.92]	[-2.94]	[-1.79]	[-4.44]	[-0.22]	[-0.80]	[-4.28]
rmw	9.56	1.21	4.77	10.86	7.81	16.06	-8.45
	[3.20]	[0.34]	[1.24]	[2.91]	[1.98]	[4.31]	[-2.44]
cma	12.67	-0.54	4.11	5.76	3.78	14.19	-5.89
	[2.98]	[-0.11]	[0.79]	[1.08]	[0.70]	[2.69]	[-1.28]
umd	-1.60	-2.31	-3.22	0.07	-1.79	-3.05	-1.96
	[-1.12]	[-1.51]	[-1.98]	[0.04]	[-1.08]	[-1.78]	[-1.45]
# months	624	624	620	620	624	620	620
$\bar{R}^2(\%)$	53	46	40	37	37	34	60

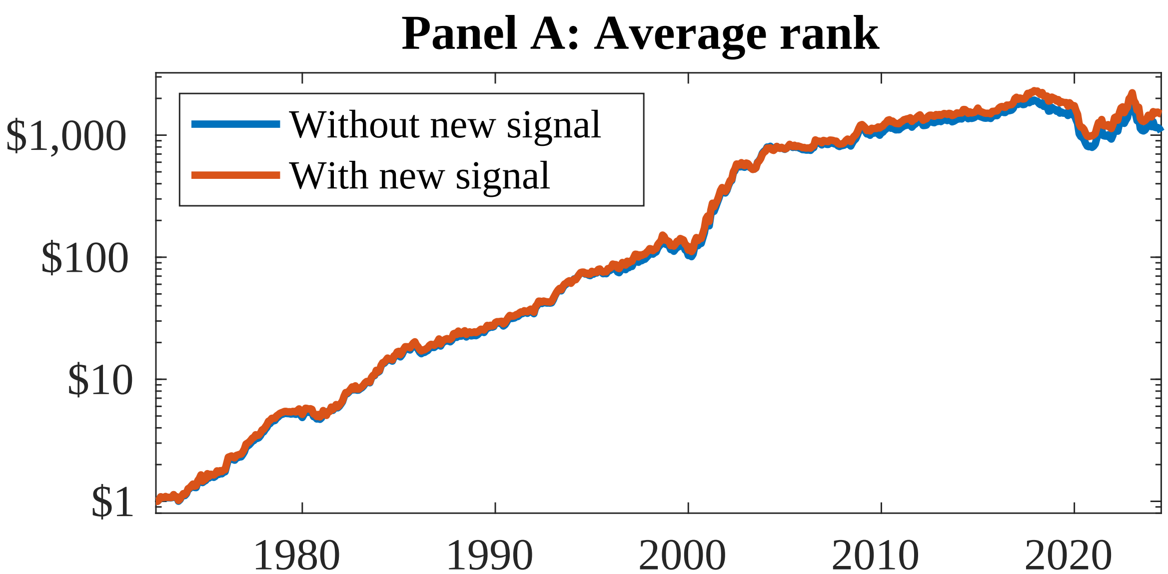


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as BDC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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